COREPRESENTABILITY OF THE FLAT AT A POINT FUNCTOR IN THE HENSELIAN ESSENTIALLY FINITE TYPE CASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex, lines 1-85) is a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project's own reference/slogan/history display-environment definitions copied mechanically from the shared preamble; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler, and extra_packages is left empty because everything needed is already in the retained preamble. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- The functor of points where a module is flat, and its invariance under a flat local extension of the target. Same chapter flat.tex, situation labelled situation-flat-at-point (statement, proof line 6792), labels lemma-flat-at-point-go-up (proof line 6816), definition-complete-devissage-at-x-algebra (proof line 6828), lemma-flat-at-point-finite (proof lines 6831 and 6840) and lemma-flat-at-point part (2) (proof line 6900). Fixes the functor F_lf being corepresented, makes it unchanged when B is replaced by a local ring flat over B, transports the problem along the finite and surjective steps of a complete devissage, and converts the corepresentation for a filtered union of ideals into finite generation.
- Existence of a complete devissage in the henselian case. Same chapter flat.tex, label lemma-existence-algebra, invoked at proof line 6819. Allows one to assume a complete devissage of N over C over A at q; this is the structural input that makes the induction on the length of the devissage possible.
- Descent of flatness along the induction step and flatness of a universally exact sequence. Same chapter flat.tex, labels lemma-induction-step (proof lines 6853 and 6874), lemma-complete-devissage-flat-finite-type-module (proof line 6871) and theorem-flattening-map (proof line 6889); Algebra, label algebra-lemma-flat-universally-injective (proof line 6885); Algebra, label algebra-lemma-unique-prime-over-localize-below and More on Morphisms, label more-morphisms-lemma-finite-morphism-single-point-in-fibre (proof lines 6836-6838). Gives that flatness propagates from M_1 to its cokernel along

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the induction step, makes the cokernel free over A once it is flat, turns the exact sequence into a universally exact one so that injectivity of the base changed map is detected on the presentation, and supplies the flattering criterion of the preceding theorem for the final corepresentation.

Source reference. Original lemma 058M: sources/stacks-chow/tags.txt, source lines 6785–6785. Navigation pointer for \ref{algebra-lemma-flat-universally-injective} used inside the counted span (line 6885); the argument that consumes it is retained in the span.

Source reference. Original lemma 00EA: sources/stacks-chow/tags.txt, source lines 527–527. Navigation pointer for \ref{algebra-lemma-unique-prime-over-localize-below} used inside the counted span (line 6836); the argument that consumes it is retained in the span.

Source reference. Original lemma 05HZ: sources/stacks-flat/pinned-original-source.tex, source lines 1710–1710. Navigation pointer for \ref{definition-complete-devissage-at-x-algebra} used inside the counted span (line 6828); the argument that consumes it is retained in the span.

Source reference. Original lemma 05I4: sources/stacks-flat/pinned-original-source.tex, source lines 2426–2426. Navigation pointer for \ref{lemma-complete-devissage-flat-finite-type-module} used inside the counted span (line 6871); the argument that consumes it is retained in the span.

Source reference. Original lemma 05I1: sources/stacks-flat/pinned-original-source.tex, source lines 1757–1757. Navigation pointer for \ref{lemma-existence-algebra} used inside the counted span (line 6819); the argument that consumes it is retained in the span.

Source reference. Original lemma 05MN: sources/stacks-flat/pinned-original-source.tex, source lines 6045–6045. Navigation pointer for \ref{lemma-flat-at-point} used inside the counted span (line 6900); the argument that consumes it is retained in the span.

Source reference. Original lemma 05P4: sources/stacks-flat/pinned-original-source.tex, source lines 6068–6068. Navigation pointer for \ref{lemma-flat-at-point-finite} used inside the counted span (line 6831); the argument that consumes it is retained in the span.

Source reference. Original lemma 05P5: sources/stacks-flat/pinned-original-source.tex, source lines 6108–6108. Navigation pointer for \ref{lemma-flat-at-point-go-up} used inside the counted span (line 6816); the argument that consumes it is retained in the span.

Source reference. Original lemma 05I3: sources/stacks-flat/pinned-original-source.tex, source lines 2356–2356. Navigation pointer for \ref{lemma-induction-step} used inside the counted span (line 6853); the argument that consumes it is retained in the span.

Source reference. Original lemma 05B8: sources/stacks-chow/tags.txt, source lines 6876–6876. Navigation pointer for \ref{more-morphisms-lemma-finite-morphism-single-point-in-fibre} used inside the counted span (line 6838); the argument that consumes it is retained in the span.

Source reference. Original lemma 05MK: sources/stacks-flat/pinned-original-source.tex, source lines 6008–6008. Navigation pointer for \ref{situation-flat-at-point} used inside the counted span (line 6792); the argument that consumes it is retained in the span.

Source reference. Original lemma 05PF: sources/stacks-flat/pinned-original-source.tex, source lines 6671–6671. Navigation pointer for \ref{theorem-flattening-map} used inside the counted span (line 6889); the argument that consumes it is retained in the span.

Theorem 0.1. *In Situation 05MK assume A is henselian, B is essentially of finite type over A, and M is a finite B-module. Then there exists an ideal $I \subset A$ such that A/I corepresents the functor F_{lf} on the category $\mathcal{C}$. In other words given a local homomorphism of local rings $\varphi : A \to A'$ with $B' = B \otimes_A A'$ and $M' = M \otimes_A A'$ the following are equivalent:*

(1) $\forall \mathfrak{q} \in V(\mathfrak{m}_{A'} B' + \mathfrak{m}_B B') \subset \mathrm{Spec}(B') : M'_{\mathfrak{q}}$ *is flat over A', and*
(2) $\varphi(I) = 0$.

If B is essentially of finite presentation over A and M of finite presentation over B, then I is a finitely generated ideal.

Proof. Choose a finite type ring map $A \to C$ and a finite C-module N and a prime $\mathfrak{q}$ of C such that $B = C_{\mathfrak{q}}$ and $M = N_{\mathfrak{q}}$. In the following, when we say "the theorem holds for $(N/C/A, \mathfrak{q})$ we mean that it holds for $(A \to B, M)$ where $B = C_{\mathfrak{q}}$ and $M = N_{\mathfrak{q}}$. By Lemma 05P5 the functor F_{lf} is unchanged if we replace B by a local ring flat over B. Hence, since A is henselian, we may apply Lemma 05I1 and assume that there exists a complete dévissage of $N/C/A$ at $\mathfrak{q}$.

Let $(A_i, B_i, M_i, \alpha_i, \mathfrak{q}_i)_{i=1,\ldots,n}$ be such a complete dévissage of $N/C/A$ at $\mathfrak{q}$. Let $\mathfrak{q}'_i \subset A_i$ be the unique prime lying over $\mathfrak{q}_i \subset B_i$ as in Definition 05HZ. Since $C \to A_1$ is surjective and $N \cong M_1$ as C-modules, we see by Lemma 05P4 it suffices to prove the theorem holds for $(M_1/A_1/A, \mathfrak{q}'_1)$. Since $B_1 \to A_1$ is finite and $\mathfrak{q}_1$ is the only prime of B_1 over $\mathfrak{q}'_1$ we see that $(A_1)_{\mathfrak{q}'_1} \to (B_1)_{\mathfrak{q}_1}$ is finite (see Algebra, Lemma 00EA or More on Morphisms, Lemma 05B8). Hence by Lemma 05P4 it suffices to prove the theorem holds for $(M_1/B_1/A, \mathfrak{q}_1)$.

At this point we may assume, by induction on the length n of the dévissage, that the theorem holds for $(M_2/B_2/A, \mathfrak{q}_2)$. (If $n = 1$, then $M_2 = 0$ which is flat over A.) Reversing the last couple of steps of the previous paragraph, using that $M_2 \cong \mathrm{Coker}(\alpha_2)$ as B_1-modules, we see that the theorem holds for $(\mathrm{Coker}(\alpha_1)/B_1/A, \mathfrak{q}_1)$.

Let A' be an object of $\mathcal{C}$. At this point we use Lemma 05I3 to see that if $(M_1 \otimes_A A')_{\mathfrak{q}'}$ is flat over A' for a prime $\mathfrak{q}'$ of $B_1 \otimes_A A'$ lying over $\mathfrak{m}_{A'}$, then $(\mathrm{Coker}(\alpha_1) \otimes_A A')_{\mathfrak{q}'}$ is flat over A'. Hence we conclude that F_{lf} is a subfunctor of the functor F'_{lf} associated to the module $\mathrm{Coker}(\alpha_1)_{\mathfrak{q}_1}$ over $(B_1)_{\mathfrak{q}_1}$. By the previous paragraph we know F'_{lf} is corepresented by A/J for some ideal $J \subset A$. Hence we may replace A by A/J and assume that $\mathrm{Coker}(\alpha_1)_{\mathfrak{q}_1}$ is flat over A.

Since $\mathrm{Coker}(\alpha_1)$ is a B_1-module for which there exist a complete dévissage of $N_1/B_1/A$ at $\mathfrak{q}_1$ and since $\mathrm{Coker}(\alpha_1)_{\mathfrak{q}_1}$ is flat over A by Lemma 05I4 we see that $\mathrm{Coker}(\alpha_1)$ is free as an A-module, in particular flat as an A-module. Hence Lemma 05I3 implies $F_{lf}(A')$ is nonempty if and only if $\alpha \otimes 1_{A'}$ is injective. Let $N_1 = \mathrm{Im}(\alpha_1) \subset M_1$ so that we have exact sequences

$$0 \to N_1 \to M_1 \to \mathrm{Coker}(\alpha_1) \to 0 \quad \text{and} \quad B_1^{\oplus r_1} \to N_1 \to 0$$

The flatness of $\mathrm{Coker}(\alpha_1)$ implies the first sequence is universally exact (see Algebra, Lemma 058M). Hence $\alpha \otimes 1_{A'}$ is injective if and only if $B_1^{\oplus r_1} \otimes_A A' \to N_1 \otimes_A A'$ is an isomorphism. Finally, Theorem 05PF applies to show this functor is corepresentable by A/I for some ideal I and we conclude F_{lf} is corepresentable by A/I also.

To prove the final statement, suppose that $A \to B$ is essentially of finite presentation and M of finite presentation over B. Let $I \subset A$ be the ideal such that F_{lf} is

corepresented by A/I. Write $I = \bigcup I_\lambda$ where I_λ ranges over the finitely generated ideals contained in I. Then, since $F_{lf}(A/I) = \{*\}$ we see that $F_{lf}(A/I_\lambda) = \{*\}$ for some λ, see Lemma 05MN part (2). Clearly this implies that $I = I_\lambda$. $\square$